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The influence of linear feature width on small mammal communities in northeastern Alberta

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Abstract

Anthropogenic linear features, such as seismic lines, pipelines, and powerlines, are widespread in the boreal forests of western Canada. Although these corridors are widely recognized as drivers of habitat loss and fragmentation, the extent of their impact is likely depending on their width. Understanding how linear feature width influences small mammal communities is important for informing more effective management and restoration efforts. We deployed remote cameras at 354 sites across a gradient of soft (vegetated) linear feature widths (3–150 m) and paired each site with forest interior control sites. We recorded twelve small mammal species. Community-level analyses showed that forest-affiliated species were associated with narrower features and forest interiors whereas open habitat species showed the opposite pattern. Species-specific models indicated that four of nine analyzed species showed significant negative responses to increasing width. No species showed a significant positive response to width. Contrary to expectations, wider features were not more likely to be occupied by early-successional species but showed reduced overall small mammal use. Our findings indicate that wide linear features may act as ecological barriers for small mammals and suggest that minimizing linear feature width and/or promoting revegetation could mitigate impacts on small mammals in the boreal forest.

Key words: boreal, linear features, seismic, pipelines, habitat fragmentation, small mammals

Introduction

Over the last two decades, energy-sector development has significantly increased across the western North American boreal forest, altering vegetation structure and composition and potentially causing habitat loss and fragmentation for some species (Venier et al. 2014; Allred et al. 2015; Pattison et al. 2016). The most extensive disturbances created by the energy sector are linear features that are used for transportation (roads), movement of energy products (powerlines and pipelines), or remote access and exploration (seismic lines; Dabros et al. 2018). In the Oil Sands Region of northeastern Alberta, linear feature density can reach up to 40 km/km^2 (Stern et al. 2018). The extensive network of linear features not only dissects large areas of the boreal forest but, in aggregate, removes considerable amounts of trees and shrubs (Dabros et al. 2022). In some cases, vegetation regrowth on linear features can be rapid (i.e., like a harvested area), while in other areas, the linear features may not regrow trees or shrubs for decades (Lee and Boutin 2006; van Rensen et al. 2015).

While the ecological impacts of roads have been extensively studied (Coffin 2007; Bennett 2017), comparatively more research is needed to understand how soft linear features (those with vegetated surfaces such as transmission lines, pipelines, and seismic lines, compared to those with non-vegetated surfaces like roads) contribute to habitat change. The magnitude and spatial extent of ecological effects associated with soft linear features are likely related to their width (Franklin et al. 2021; Kalukapuge et al. 2024). Seismic lines are the narrowest feature, typically 2–8 m wide, while pipelines range from 8 to 30 m, powerlines tend to exceed 30 m, and shared pipeline and transmission line corridors can reach up to 80–130 m wide. This variation in width raises the question about when linear features are perceived as forest discontinuities by animals leading to changes in their habitat use. Narrow linear features may not even be perceived as a gap or edge by some species (Machtans 2006; Bayne and Dale 2011). Conversely, features such as pipelines and powerlines clear wider sections of the forest, increasing the potential to be perceived as a gap by wildlife and cause habitat loss due to the more divergent physical conditions between the clearing and the adjacent forest (e.g., humidity or sun exposure, Ries et al. 2004). Wider linear features may also be less likely to support full forest regeneration because of management practices (Government of Canada 2019; Government of Alberta 2023). Powerlines right-of-way, for instance, are typically maintained in an early-successional regeneration stage dominated by shrubs or graminoids to prevent trees from falling onto electrical infrastructure (Government of Alberta 2018*a*). Likewise, pipelines, although they exhibit more variable levels of vegetation recovery, are regularly maintained and typically retain

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a truck-width trail actively used by vehicles (Government of Alberta 2018*b*; Canada Energy Regulator 2025). Conversely, narrower features or those with vegetation similar to the surrounding forest may permit some level of habitat continuity. Wildlife in the boreal forest have shown a wide range of responses to soft linear features including, increased movement (e.g., wolf (*Canis lupus* Linnaeus, 1758), Latham et al. 2011; Dickie et al. 2017), increased diversity (e.g., butterflies and birds Riva et al. 2018; Kalukapuge et al. 2024), increased mortality rates (e.g., caribou (*Rangifer tarandus caribou* (Gmelin, 1788), COSEWIC 2014), and reduced abundance (e.g., Canada warbler (*Cardellina canadensis* (Linnaeus, 1766), Gregoire et al. 2022). However, almost all of these studies have focused on seismic lines, and few have looked at how the width of linear features influences wildlife generally (but see Kalukapuge et al. 2024).

Small mammals are a group of species particularly suited for investigating whether linear feature width is important. Species within this group vary widely with respect to the vegetation conditions they select for with some boreal species preferring grass and others closed canopy forests (Galindo and Krebs 1985; Bowers et al. 1996; Lemaitre et al. 2010; Vanderwel et al. 2010). Additionally, the limited mobility and relatively small home ranges of many small mammals may make them more sensitive to habitat loss/gain as a consequence of the line clearing. In fact, thresholds may arise when the width of a linear feature exceeds the circumference of a typical individual's home range size (Blair 1942; Morris 1955; Getz 1961; Wrigley et al. 1979; Tisell 2018). For species that require forested conditions, this may represent loss of habitat, whereas those preferring grass and shrubs may have the potential to establish home ranges entirely within the footprint of a linear feature. Previous research suggests that wide linear features such as pipelines prompt shifts in small mammal communities from forest-dwelling species (Southern red-backed voles (*Myodes gapperi* (Vigors, 1830)) to those adapted to open habitats (meadow voles (*Microtus drummondii* (Audubon and Bachman, 1854)); Darling et al. 2019). Research on power lines has shown a decline in Southern red-backed voles on the right-of-way, which tends to be used instead by generalist species, such as deer mice (*Peromyscus maniculatus* (Wagner, 1844)) and least chipmunks (*Neotamias minimus* (Bachman, 1839), Fortin 2008; Storm and Choate 2012). Data on the response of other small mammal species to seismic lines is more limited, although martens (*Martes americana* (Turton, 1806)) avoided lines that were open and wider than 2 m (Tigner et al. 2015).

We investigated the relationship between linear feature width (of seismic lines, pipelines and powerlines) and probability of use by small mammals in northeast Alberta. Using remote wildlife cameras, we detected a range of herbivores and carnivores ranging from voles and shrews to martens and weasels. We hypothesized that individuals from each species would show species-specific responses to line width based on their ecological requirements. We predicted that (1) as the linear feature gets wider, we would observe a lower probability of use by forest-affiliated species and (2) an increased probability of use by early-successional species. We also predicted that (3) generalist species' probability of use wouldn't change with linear feature width and should be similar across all of the linear feature widths considered.

Materials and methods

Study area

Our study was conducted in the southern portion of Alberta's Lower Athabasca region, in an area ranging from Cold Lake (54.4642°N, 110.1732°W) and Wolf Lake (54.6689°N, −110.9724°W) to the South, Wandering River (55.2014°N, 112.4652°W) to the West, and Anzac (56.4489°N, 111.0368°W) to the North, encompassing both oil leases and crown land (Fig. 1). The study area was characterized by a relatively flat terrain consisting of depressed areas (lowlands) transitioning into more elevated areas (uplands). Upland areas were typically deciduous forests dominated by the trembling aspen (*Populus tremuloides* Michx, 1803) or mixedwood forests and contained varying percentages of white spruce (*Picea glauca* (Moench) Voss, 1907) and aspen. Lowland habitats were dominated by treed fens, where black spruce (*Picea mariana* (Mill.) Britton, Sterns & Poggenb, 1888) and tamarack (*Larix laricina* (Du Roi) K. Koch, 1873) are the most common species. Transitional areas were typically coniferous forests characterized by various percentages of black spruce cover alternated with jack pine (*Pinus banksiana* Lamb, 1803) trees.

Sampling design and data collection

Between May and August 2023, we sampled 354 sites in both upland and lowland forests. For sampling purposes, we stratified by linear feature width when selecting sites. Linear feature width ranged from 3 to 150 m. For site selection, we divided linear features into three categories (1–5, 6–10, and 10–150 m), which reflect the three main types of linear features characterized by vegetated on-footprint surface: low-impact seismic, conventional seismic, and pipelines/transmission lines. We sampled a minimum of 15 replicates per width category in each of three ecotypes: upland deciduous, mixedwood with an approximately equal representation of deciduous and coniferous species, and lowland coniferous, including treed fens and treed bogs (see Table 1 for details on the number of replicates per treatment and ecotypes). The linear features sampled were randomly selected to be at least 100 m and no more than 5 km from the nearest road. Replicates were selected to maintain a minimum spacing of 100 m between each other and any other linear feature (except for low-impact seismic lines because they are never more than 50 m apart). As the time since the last disturbance widely varied between linear features, we tried to balance our sites by purposefully sampling linear features at different regeneration stages, avoiding those that were recently disturbed (within the last 3–4 years) and completely barren. Similarly, because the linear features in our study area were known to experience different levels of human activity related to the recreational use of off-highway vehicles (Franceschini, pers. obs.), we avoided seismic lines that were intensively used as off-highway vehicle (OHV) trails or winter roads.

Fig. 1. Camera trap sampling locations across the study area. The Alberta Oil Sands Region, including the Athabasca, Cold Lake, and Peace River areas is outlined in different colors (Government of Alberta 2017). Individual locations are shown as blue points overlaid on a density heatmap, where warmer colors indicate higher concentrations of sampled sites within a 10 km radius. Provincial boundaries are outlined for geographic reference (Statistics Canada 2023). Coordinates are reported in decimal degrees using the North American Datum 1983 (NAD83) geographic coordinate system.

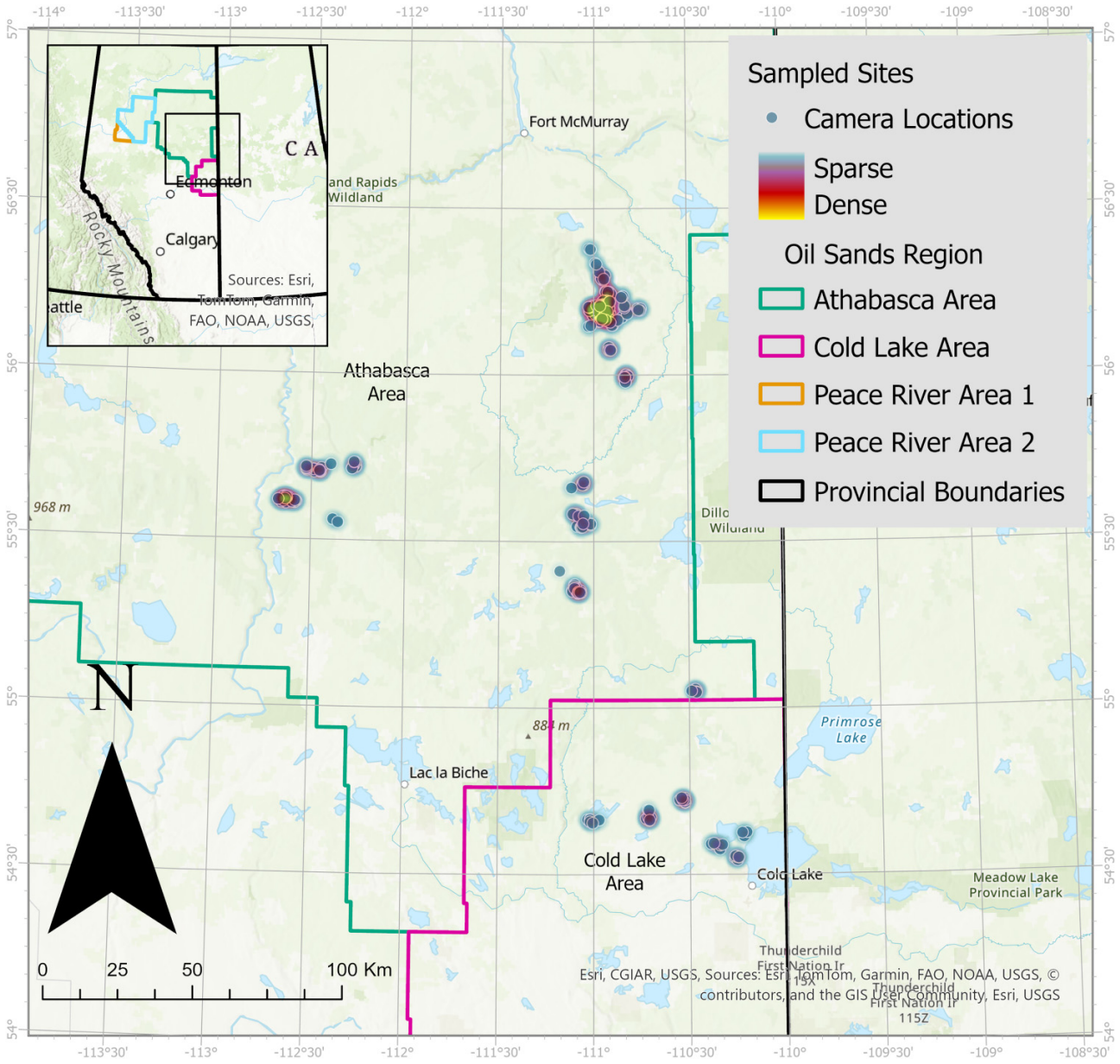


Table 1. Number of sampled sites ($n = 354$) divided by treatment and ecotype.

Treatment	Upland deciduous forest	Mixedwood forest	Lowland coniferous forest
Low-impact seismic	20 (17)	18 (16)	18 (16)
Control	20 (18)	18 (16)	18 (17)
Conventional seismic	21 (16)	20 (16)	20 (17)
Control	21 (18)	20 (16)	20 (17)
Pipeline/transmission line	20 (15)	20 (18)	20 (15)
Control	20 (19)	20 (18)	20 (17)

Note: Numbers in parentheses are the number of sampled sites after the failed cameras were removed ($n = 302$).

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We used camera traps (models Reconyx PC900 and Reconyx Hyperfire 2) with a modified focal length (approx. 25 cm) to survey the small mammal community. As a general framework, we followed the method outlined by Gracanin et al. (2019) with some adjustments (see below). The camera was placed on the ground at the center of the line and strapped to a wooden stake pounded into the ground to keep the camera in place. The area in front of the camera up to 25 cm was cleared of vegetation or debris to produce an unobstructed view of the animal and reduce false triggering (Kelly and Holub 2008; Glen et al. 2013). We placed a baited PVC pipe 25 cm from the camera and fixed it to the ground using metal pegs to prevent it from moving due to animal activity. The pipe was 7 cm wide and 40 cm long; one entrance was previously closed with a lid while the other faced the camera. The cameras remained active for four consecutive nights before being retrieved. Because we only baited during the deployment, we used a higher quantity of bait (approximately 100 g of a mix of peanut butter, oats, and honey) to ensure that it would not run out and to slow down the drying process (which was further decreased by the cover offered by the pipe). For each sampled line point, we also sampled a control site located in the adjacent forest, 50 m from the edge of the linear feature. In the case of low-impact seismic lines, the dense grid networks prevented us from keeping this spacing, which was reduced to 25 m from the edge of the line. The site we used for the control needed to be undisturbed and at least 50 m from any other footprint. We chose 50 m because previous studies on small mammals showed no significant edge effects at this distance from the edge (Darling et al. 2019). At each site, we collected data on the date and time of deployment/retrieval, ecotype, and line width.

All of the work done in this study was done according to the guidelines for the use of mammals in research by the American Society of Mammalogy (Sikes and Gannon 2011). This work was covered under an institutional animal care protocol approved by the University of Alberta Animal Use Protocol Committee under permit number AUP00004093 and under the authority of collection permits from the province of Alberta through the Alberta government online permitting and clearing center. Wildlife Research Permit and Collection Licence 23-377.

Statistical analysis

We processed the camera trap data using WildTrax, a platform for managing and analyzing acoustic and camera data. The camera's short focal length and the extended time animals spent in view allowed us to identify most of the encounters at a species level based on distinctive physical features. The only exception was for shrews, which are difficult to differentiate due to the similar size, coat color, and body shape across different shrew species. Typically, species-level identification of shrews requires dental examination, so in our case, we grouped all shrew encounters under "*Sorex* spp." as the shrew species that can be found in our study area are: the masked shrew (*Sorex cinereus* Kerr, 1792), the American pygmy shrew (*Sorex hoyi* Baird, 1857), the prairie shrew (*Sorex haydeni* Baird, 1857), the arctic shrew (*Sorex arcticus* Kerr, 1792) and the American water shrew (*Sorex palustris* Richardson, 1828). In rare cases (<1% of images), where only a small part of the animal was visible but blurred, making species identification impossible, we categorized the encounter as "unidentified."

The raw camera trap data containing the tagged pictures was reformatted to create a separate entry for the starting and ending time and date, which was retrieved from the camera trap metadata. We then removed sites where either bear interference or camera malfunction occurred ($n = 52$, Table 1). The data was further filtered to include only detections from the 2nd to the 94th hour post-deployment. Using this filtered dataset, we created a presence–absence matrix for each species. We used the matrix to calculate the species-specific naïve occupancy (ψ), which is computed as the number of sites the species was detected divided by all sites sampled ($n = 302$), and we filtered out the dataset to only include species with a $\psi \geq 0.05$ to avoid including rare species for which model inference was not possible due to low sample size. Before fitting our models, we transformed the date of camera deployment into day of the year to account for seasonal patterns in resource distribution, changing densities, or differential species activity patterns that might affect the probability of use. Similarly, we grouped the forest types into three ecotypes: upland deciduous, mixedwood, and lowland coniferous, to account for a higher or lower probability of use due to species-specific vegetation preferences. We then explored the data to identify any outliers and check for collinearity between covariates. We decided to log-transform width to reduce skewness and we used the formula: $\log(x + 1)$ due to the presence of zeros. We also scaled log-width and day of the year to improve model fit.

Community level analyses

To assess the influence of environmental variables on species composition, we employed redundancy analysis (RDA) using the "*vegan*" package in R (Oksanen 2009). RDA is a constrained ordination method particularly suited for presence–absence or abundance data when the aim is to relate species composition directly to measured environmental gradients (Legendre and Legendre 2012). This method is appropriate for our hypothesis, which posits that linear feature width and associated environmental factors structure community composition in predictable ways. RDA is especially suitable when assessing non-directional variation in community composition (i.e., beta diversity sensu Anderson et al. 2011), as it allows quantification and partitioning of the explained variation in multivariate species data in relation to environmental predictors.

A Hellinger-transformation was used on the presence–absence community matrix to meet the Euclidean distance assumption required by RDA (Legendre and Gallagher 2001). The explanatory variables included the main predictor (log-transformed width) and the controlling variables (ecotype (classified as deciduous, mixedwood, or lowland coniferous) and day of the year). Sites were grouped by location (Forest interior or Line), and 95% confidence ellipses were drawn around each group to visualize differences in species com-

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position. A global ANOVA test was performed to determine the significance of the model (9999 permutations), followed by marginal tests to assess the effect of each predictor after accounting for the others. For this test, we included in the model a spatial variable using latitude and longitude as covariates to control for unique spatial variation unaccounted by the design factor and covariates. The adjusted R^2 value was calculated to evaluate model fit, and variation partitioning was performed to quantify the unique contributions of the predictors to the total explained variance (Oksanen et al. 2013).

Finally, we divided the species based on their expected vegetation affiliation related to their typical use of older forests versus other vegetation conditions. These three groups were: (1) early-successional species, (2) generalists species, and (3) forest-affiliated species. This classification was informed by life history traits described in Soria et al. (2021). To evaluate how species traits interact with environmental gradients, we conducted a fourth-corner analysis (Dray and Legendre 2008; ter Braak et al. 2012) using the "*ade4*" package in R (Dray and Dufour 2007). This method is appropriate when testing whether functional traits mediate species responses to environmental gradients, in our case, whether forest affiliation influences species' associations with linear feature width. The analysis integrates three matrices: environmental variables (X), species composition (L), and traits (Q), and quantifies trait–environment associations via permutation-based significance testing. It also estimates a Pearson correlation coefficient for quantitative (e.g., linear feature width) and PseudoF statistic for discrete (e.g., ecotype) environmental variables. This allowed us to formally assess whether trait–environment associations (e.g., stronger responses by forest-affiliated species) aligned with the observed compositional patterns identified in the RDA.

Species-specific analyses

Generalized linear mixed models (GLMMs) were employed for species-specific responses to log-width, allowing for covariate control, spatial random effects, and correction for autocorrelation, offering robust inference at the individual species level. These methods are appropriate and complementary to address our gradient-based hypotheses across community, trait, and species levels. We built GLMMs using species presence/absence data and a binomial error distribution in the "*glmmTMB*" package in R (Brooks et al. 2017). For each species, we fitted log-width into a baseline model that would take into account the portion of variability due to ecotype and day of the year. Both line and control points were included in these analyses, with control points treated as having a line width of 0. To account for local spatial patterns, we included each site pair (control and line point) as a random effect. To account for broader-scale, non-modeled spatially structured environmental patterns, we used a residual autocovariate model (Crase et al. 2012). We fitted the residuals of the GLMM model using a smoothing term for latitude and longitude as covariates (i.e., a trend-surface model) within a generalized additive model (GAM) with a Gaussian distribution (mgcv package; Wood 2017). The fitted values from the spatial GAM were then scaled and incorporated as a fixed effect in the final GLMM to account for spatial autocorrelation and large-scale trends (Crase et al. 2012). The "*DHARMa*" package in R was used to assess model fit, checking residuals for traces of model misspecification. For species with fewer detections, if model diagnostics indicated a misfit, we attributed it to the poor predictive ability of the original covariates. This likely resulted from an excessive number of parameters relative to the limited number of detection sites available for estimating covariate effects. In these cases, we simplified the model by reducing the number of controlling covariates (removing day of the year and/or ecotype as needed), while retaining log-width as the primary predictor. We then re-ran the residual autocovariate model, re-fitted the GLMM with the updated spatial covariate, and reassessed the model fit. Fixed effect estimates and 95% confidence intervals were computed using Wald intervals. To estimate quantitatively the unique contribution of each predictor to the total variance explained by the fixed effects, we used the "*glmm.hp*" R package. To evaluate whether species responses to log-width were better captured by nonlinear patterns, we fit an additional set of models including a quadratic term for non-transformed linear feature width. For each species, we updated the initial binomial GLMM to include a squared term for the scaled covariate linear feature width. We then compared model performance between the log-transformed and quadratic formulations using Akaike's Information Criterion (AIC). For each species, we calculated the AIC of the log-transformed model and the corresponding AIC of the quadratic model, and computed the ΔAIC (AIC_log—AIC_quadratic) to assess model improvement. A positive ΔAIC (>2) indicated support for the inclusion of the quadratic term. All analyses and visualizations were performed in R version 4.4.1.

Results

Overall, the 302 camera traps accumulated 1306 camera days (mean per site 4.3) between 24 May to 20 August 2023. Sampling yielded 2449 229 images of which 540 262 were of wild mammals, 15 143 were humans (setup/pickup), 1891 206 were false triggers, and 2618 were unidentifiable. Unidentifiable were images where the subject appears blurry, overexposed, or outside the flash range, making it impossible to identify the species. Twenty-one species of animals were recorded, 15 of which were mammals (Table 3). The list of mammal species recorded and the proportion of sites where they were detected (detection-naïve Ψ) are reported in Table 2.

Although the number of control (forest) and treatment (line) sites was relatively balanced (157 forest vs. 146 line), the number of sites with no animal detections was higher in the treatment sites (forest: 18, line: 42). A Fisher's exact test indicated that the likelihood of recording zero animal detections was significantly greater at line sites ($P < 0.001$), with an average 32% higher probability of observing no animal detections compared to forest sites.

Table 2. Checklist of camera-trapped mammals in the study area, ordered by decreasing number of events using a 1-day independent detection interval.

Taxonomic group	Latin name	Common name	Habitat affiliation	Number of events	Naïve Ψ
Rodentia	*Myodes gapperi*	Southern Red-backed Vole	2	282	0.42
Eulipotyphla	*Sorex* spp.	Shrews	2	260	0.45
Rodentia	*Tamiasciurus hudsonicus*	American Red Squirrel	3	129	0.24
Rodentia	*Peromyscus maniculatus*	Deer Mouse	2	122	0.16
Rodentia	*Microtus drummondii*	Meadow Vole	1	46	0.09
Lagomorpha	*Lepus americana*	Snowshoe Hare	2	37	0.09
Carnivora	*Martes americana*	Marten	3	30	0.07
Rodentia	*Zapus princeps*	Western Jumping Mouse	1	22	0.06
Carnivora	*Mustela erminea*	Short-tailed Weasel	2	17	0.05
Carnivora	*Ursus americanus* Pallas, 1780	American Black Bear	2	8	0.02
Rodentia	*Glaucomys sabrinus* (G. K. Shaw, 1801)	Northern Flying Squirrel	2	5	0.009
Rodentia	*Marmota monax* (Linnaeus, 1758)	Groundhog	1	1	0.003
Carnivora	*Procyon lotor* (Linnaeus, 1758)	Common Raccoon	2	1	0.003
Carnivora	*Canis lupus* familiaris Linnaeus, 1758	Domestic Dog	2	1	0.003
Ungulates	*Odocoileus virginianus* (E. A. W. von Zimmermann, 1780)	White-tailed Deer	2	1	0.003

Note: The vegetation affiliation is represented as: early-successional species (1), generalist species (2), forest-affiliated species (3). Naïve occupancy (ψ) is computed as the number of sites the species was detected divided by all sites sampled ($n = 302$).

Community level analyses

Overall, the RDA explained 11.8% of the total variation in species composition ($R^2 = 0.106$, Fig. 2). When we accounted for spatial patterns, the variance partitioning analyses explained 13.4% of the variation ($R^2 = 0.116$). Sites were positioned along RDA axis 2 (explained variance = 18.03%), which captured the gradient in log width, with species composition differing between narrower and wider features. The environmental vectors (blue arrows) show that log-width tended to be positively associated with species like the meadow vole (*Microtus drummondii* (Audubon & Bachman, 1854)) and the western jumping mouse (*Zapus princeps* Allen, 1893), both of which are early-successional species. In contrast, forest-dependent species such as Southern red-backed vole (*Myodes gapperi* (Vigors, 1830) = *Clethrionomys gapperi* (Vigors, 1830)) and American red squirrel (*Tamiasciurus hudsonicus*) were negatively associated with log-width and more strongly associated with the forest interior points (green), which cluster on the opposite side of the width vector.

In addition to log-width, day of the year contributed to species variation along RDA axis 1 (explained variance = 65%), suggesting some seasonal shifts in species presence, particularly for *Sorex* spp. The ecotype also explained some variation but was secondary to the effect of log-width. The marginal tests also showed that each predictor (day of the year, ecotypes and the spatial term comprising longitude and latitude) had a statistically significant contribution in explaining the variance ($p < 0.001$). Log-width explained 2.2% of the total variance ($F = 8.399$, $p < 0.001$).

The fourth-corner analysis revealed a statistically significant relationship between forest affiliation (across all categories) and log-width ($r = -0.226$, $p = 0.023$), indicating that forest-affiliated species (forest affiliation = 3) occur significantly more on narrower linear features, whereas the occurrence of open habitat species (forest affiliation = 1) is higher in wider linear features. No significant associations were detected for day of the year ($r = -0.144$, $p = 0.506$) or ecotype ($F = 1.948$, $p = 0.445$), suggesting that neither seasonal timing nor categorical vegetation type showed strong trait–environment alignment in this analysis.

Species-specific analyses

The results from the single-species binomial GLMMs indicated that log-width can affect the use of linear features by small mammals (Table 3). Log-width was a significant factor in the occurrence of four of the nine small mammal species analyzed: the short-tailed weasel (*Mustela erminea* Linnaeus, 1758), southern red-backed vole (*Myodes gapperi*), red squirrel (*Tamiasciurus hudsonicus* (Erxleben, 1777)), and deer mouse (*Peromyscus maniculatus*). A detailed summary of each species-specific model is provided in the Supplementary Material Table S1.

Both the southern red-backed vole and the American red squirrel exhibited a significant negative association with log-width (see Table 3), suggesting a consistent negative response to wider linear feature for both species (Fig. 3). In the red-backed vole model, the full model explained 43% of the variance, with log-width accounting for 27%. For the red squirrel, the fixed effects explained 20% of the variance, with log-width alone contributing 11%.

The deer mouse and short-tailed weasel also showed a significant negative association with log-width (Fig. 3) and the largest negative slope was found for the latter species (Table 3). However, the relatively wide confidence intervals on the slope parameter suggests some uncertainty in the relationship for both species. In the case of the deer mouse, log-width accounted for 2% of the total variance explained, while the full model explained 27%. For the short tailed weasel, the

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Fig. 2. Redundancy analysis (RDA) biplot illustrating the relationship between small mammal species composition and environmental variables across sites differing in linear feature width. Points represent individual sampling sites, colored by their location (green = Forest interior, orange = Line), with 95% confidence ellipses around each point location. Species names are shown in red at their ordination centroids. Environmental vectors (blue arrows) show the strength and direction of relationships between species composition and the explanatory variables: log-transformed feature width (Log Width), day of the year (Date), and ecotype (Deciduous (used as the intercept), Mixed wood, and Lowland). The RDA explained 11.8% of the total variation in species composition.

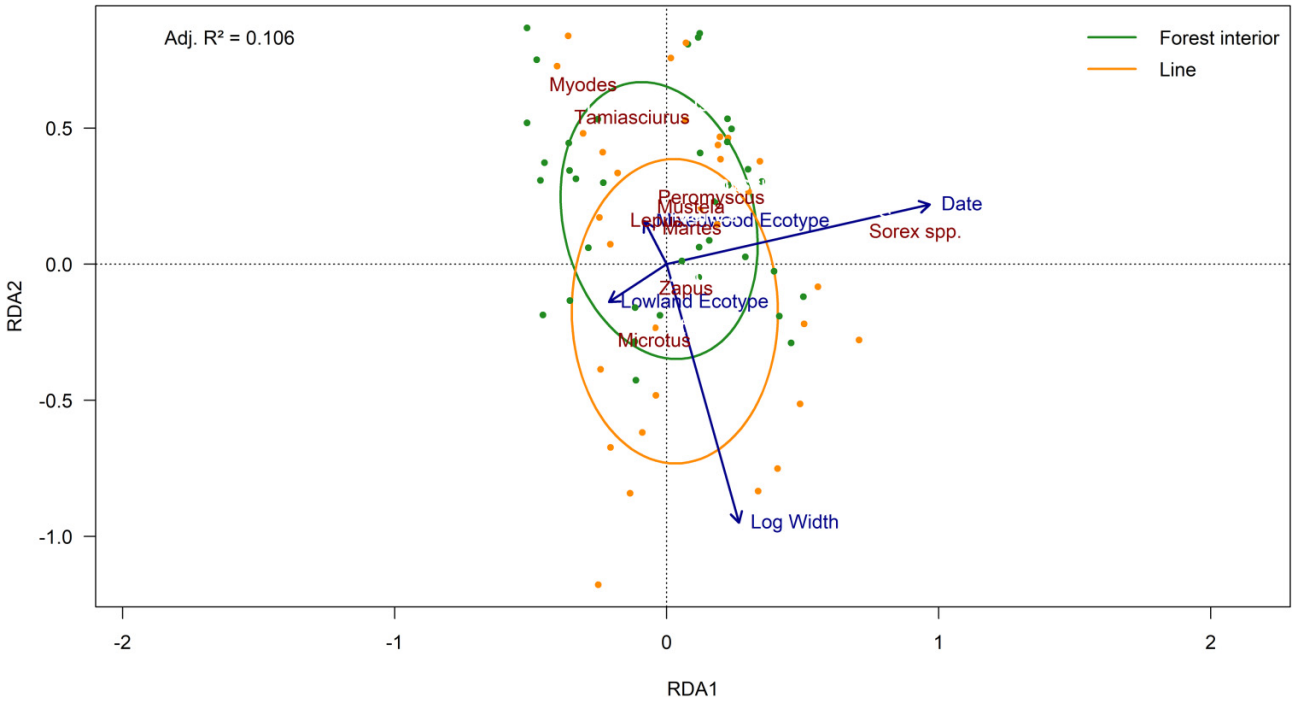


Table 3. Estimated effect of linear feature width (log-transformed and scaled) on species presence, based on generalized linear mixed models (GLMMs) for each species.

Species	Estimate	Lower CI (95%)	Upper CI (95%)	*P*
Red-backed Vole (*Myodes gapperi*)	−0.736	−1.020	−0.451	<0.001
American Red Squirrel (*Tamiasciurus hudsonicus*)	−0.710	−1.097	−0.323	<0.001
Deer Mouse (*Peromyscus maniculatus*)	−0.555	−1.021	−0.088	0.020
Snowshoe Hare (*Lepus americanus*)	−0.471	−0.943	0.001	0.051
Short-tailed Weasel (*Mustela erminea*)	−1.121	−2.078	−0.163	0.022
Sorex spp.	−0.190	−0.472	0.092	0.458
American Marten (*Martes americana*)	−0.160	−0.695	0.375	0.558
Meadow Vole (*Microtus drummondii*)	0.209	−0.167	0.585	0.275
Western Jumping Mouse (*Zapus princeps*)	0.627	−0.314	1.568	0.122

Note: The table reports the species' common and scientific names, along with the estimated effect size (β), 95% Confidence Interval (CI Lower/CI Upper), and p for log-transformed and scaled linear feature width.

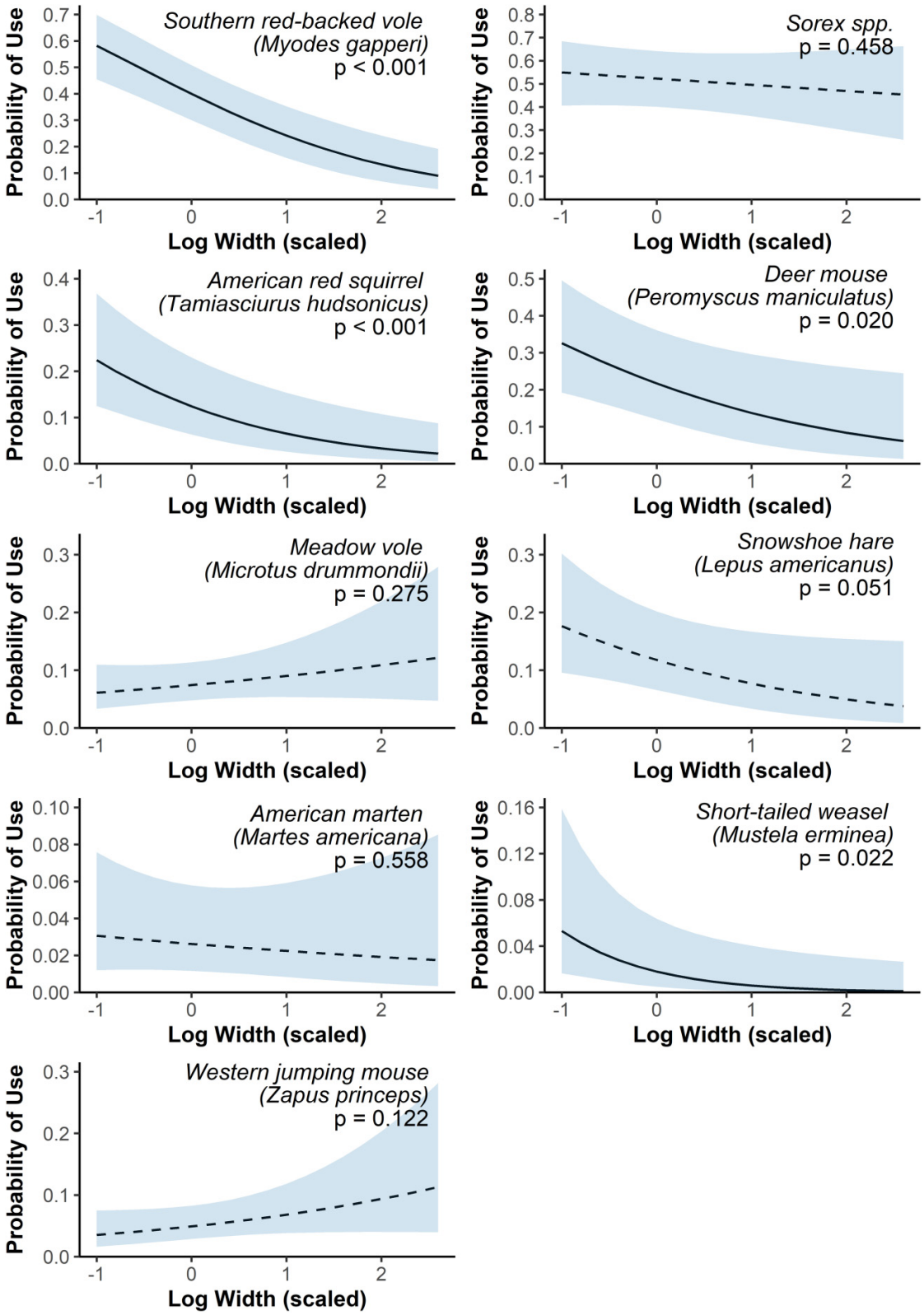
fixed effects explained 46% of the total variance, with width contributing 24%.

The model for the snowshoe hare (*Lepus americanus* Erxleben, 1777) indicated a possible negative association with log-width although the 95% confidence interval overlapped zero (Table 3, Fig. 3). Log-width accounted for 5% of the variance in snowshoe hare presence, with the full model only explaining 13%. For American marten, *Sorex* spp., meadow vole and western jumping mouse, our analysis provided no evidence for an association between probability of use of these species and log-width (Fig. 3) as the 95% confidence intervals of these relationships largely overlapped zero (Table 3).

When testing for the quadratic effect of width we found no evidence that adding a quadratic term improved model fit for any of the nine species. In all cases, the difference in AIC between the log-width and quadratic models was less than 2 (ΔAIC < 2), indicating that the more complex quadratic models were not meaningfully better supported. This suggests that no species preferred intermediate width linear features. AIC comparison between log-width and quadratic models is provided in the Supplementary Material Table S2.

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Fig. 3. Relationships between the predicted probability of habitat use by each species and linear feature width (log-transformed and scaled). Predictions are from binomial generalized linear mixed models fit separately for each species, controlling for ecotype, day of the year, and spatial autocorrelation where applicable. Solid lines represent significant relationships ($p \leq 0.05$), and dashed lines represent non-significant relationships ($p > 0.05$). Shaded ribbons indicate 95% confidence intervals. Species names are shown in each panel with associated *p*-values for the effect of seismic line width.



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Discussion

Our findings contribute insights into the complex interplay between small mammals and anthropogenic disturbance in boreal ecosystems. The results from the community and species-specific models highlight that linear features impact small mammal use of the landscape. Linear feature width was a significant predictor for four of the nine species analyzed: the Southern red-backed vole, American red, deer mouse, and short-tailed weasel. The relationship between species use and line width was consistently negative or neutral for every species we had sufficient data to model, although the pattern was most pronounced among forest-affiliated species. Overall, wider linear features, such as pipelines and power lines, seem to reduce the probability of use by many species, and there was no strong evidence of replacement by other species. Rather than a turnover from forest-affiliated species to those adapted to open habitats, our study suggests that these areas are used less overall by all species. Species that tend to use areas of the boreal forest dominated by shrubs or grass did not show significant increases in the use of wide linear features, resulting in a general reduction of species activity as linear features became wider.

The Southern red-backed vole and the red squirrel, which are typically associated with forested environments, exhibited the most significant negative response to linear feature width. This pattern was confirmed in the fourth-corner analysis, which showed a significant negative correlation between linear feature width and forest affiliation. The results support our first prediction that forest-affiliated species would show reduced use of wider linear features. These findings align with previous studies, such as Darling et al. (2019), which reported a decline in Southern red-backed vole abundance on pipelines and near pipeline edges. Both these species depend heavily on dense canopy cover and structurally complex habitats for food and shelter (Riege 1991; Goheen and Swihart 2005; Tisell et al. 2023).The Southern red-backed vole, in particular, relies on well-decayed coarse woody debris (CWD) and dense canopy cover (Fauteux et al. 2012; Fauteux et al. 2013). Anthropogenic linear clearings have a simplified vegetation structure and lower CWD availability compared to forest interiors. The open canopies of these areas also alter microclimatic conditions and may increase perceived predation risk (Moses and Boutin 2001; Carthew et al. 2013; Dabros et al. 2018; Dabros and Higgins 2023). Together, these changes reduce the suitability of these habitats for forest-dwelling rodents (Darling et al. 2019). Similar patterns of avoidance have been observed following clear-cutting and wildfire, where Southern red-backed vole populations often decline sharply in response to the sudden loss of canopy cover and microhabitat complexity (Sullivan et al. 1999; Zwolak 2009). For example, Zwolak (2009) found that red-backed voles were almost entirely absent from recently burned forests, only returning as ground cover and coarse woody debris accumulated over time. Similarly, voles tend to avoid recently clear-cut areas until sufficient regeneration of understory vegetation and coarse woody material occurs (Sullivan et al. 2017). However, post-fire forests may recover structural complexity more rapidly than linear features, particularly through the recruitment of standing deadwood and rapid regrowth of dense shrub layers, which can provide some suitable habitat for voles within a decade (Zwolak 2009). In contrast, linear features often remain in an early-successional state for prolonged periods, particularly in the boreal forest, where regeneration is slow (van Rensen et al. 2015). As a result, the impact of linear disturbances on forest-dependent species like the Southern red-backed vole may be more persistent compared to natural disturbances like fire, especially when linear corridors are maintained for ongoing industrial use.

American martens, contrary to our prediction, did not show a significant preference for forest interiors in this study, a result that differs from previous research demonstrating their preference for forest interiors over seismic lines (Tigner et al. 2015). This discrepancy likely stems from imperfect detection, as martens are elusive and may not consistently be attracted by the bait used in our study. Tigner et al. (2015), for example, used a more specific bait to attract carnivores. Given their elusive nature and the limitations of our detection methods, we think our result for marten may not reflect their actual response to linear features and instead is a consequence of imperfect and low detection probabilities.

In contrast to our second prediction, early-successional species did not show a significant increase in the use of wider features. This was particularly surprising for the meadow vole, a species often associated with grassy, open habitats (Bowers et al. 1996). This contrasts with previous live-trapping studies in northwest Alberta and the southern North Western Territories, where meadow voles were much more abundant on wide pipelines and powerlines right-of-way (Darling et al. 2019). In Darling's study areas, linear feature density and well sites were far less abundant than in our study area. Thus, the difference we observed between studies could reflect the complex interplay of local and landscape-scale effects. Although meadow voles are expected locally to select for open areas with more grass or shrubs, dispersal behaviors at larger scales might lead them to spill over into forest edges and interiors, in areas with high overall disturbance. Lin and Batzli (2001) and Oatway and Morris (2007) discussed how broader landscape connectivity and competition can shape vole distributions, even overriding local habitat preferences. This blending of scale-dependent effects may introduce noise that complicates local patterns of habitat use, underscoring the need for integrated analyses that consider broader landscape contexts. In our study area, the dense network of linear and polygonal features creates a highly heterogeneous landscape where the cumulative effects of different disturbance types at a broader scale might impact what we observe on a local scale. We hypothesize that grassy open patches like well sites or wide linear features might serve as a source for the dispersal of local populations of meadow voles with spillover into the surrounding forest. The magnitude of this spill-over is predicted to depend on the total energy footprint in the overall area. We are currently collecting data to test this hypothesis.

Similarly, the western jumping mouse is an early-successional species, typically associated with moist, herbaceous habitats with dense ground vegetation, which can be promoted along the edges of wider linear features where in-

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creased light penetration enhances vegetation growth (Hart et al. 2004). However, we didn't find a significant relationship with width. This may be due to the heterogeneity of vegetation on the sampled linear features and the small sample size we had for this species (only detected at 6% of the sites). Similar uncertainty has been documented in other studies examining small mammal responses to linear disturbances, where species-specific habitat preferences and microhabitat heterogeneity can obscure clear patterns (Fahrig 2003; Laurance et al. 2009).

Our third prediction that generalist species would exhibit consistent use across the linear feature gradient, regardless of linear feature width, was not fully supported. Only the snowshoe hare and the shrews showed a neutral response with regard to width, the deer mouse and the short-tailed weasel, were both negatively affected by linear feature width, although the estimated slope varied. For snowshoe hares, their neutral response may reflect their affinity for edge environments where cover and foraging opportunities coexist. While they are often associated with dense understory and shrub cover, hares are also known to exploit edge habitats and early-successional vegetation for feeding, particularly where browses are abundant (Pattison et al. 2020). In our study, the placement of camera traps in the center of the linear feature may have reduced the likelihood of detecting hares if they preferentially use edge zones. However, the lack of a negative trend suggests hares do use wider linear features. Similarly, shrews didn't show a significant response to linear feature width. Species within the genus *Sorex* can tolerate and even thrive in disturbed microhabitats. Studies in temperate and boreal forests have observed equal or higher shrew capture rates in disturbed or early-successional sites compared to mature forest (Panzacchi et al. 2010; Čepelka et al. 2011). For example, a broad meta-analysis in Europe found that common shrews (*Sorex araneus* Linnaeus, 1758) increased in abundance after clear-cut logging (in temperate forests) and showed no significant change in boreal forest clear-cuts due to their opportunistic traits and flexible habitat use (Michałand Rafał2014). In North America, other species of the genus *Sorex* are known to colonize burns and clear-cuts quickly once some ground cover returns, reaching high densities in young post-disturbance stands (Zwolak and Foresman 2007). Such findings confirm our results that *Sorex* shrews are likely not affected by linear features.

The negative response of deer mice to linear feature width was particularly unexpected, given their generalist nature, which allows them to thrive also in disturbed or early-successional habitats (Suzuki and Hayes 2003; Fuller et al. 2004) where they can benefit from reduced competition from forest specialists (Fisher and Wilkinson 2005; Zwolak 2009). In clearcuts, for example, deer mice often reach peak abundance within the first few years following harvest (Ramirez and Hornocker 1981; Martell 1983; Kirkland 1990). However, avoidance of linear features by deer mice has also been reported in some contexts. Darling et al. (2019) observed lower deer mouse abundance along pipelines and suggested that this could be linked to insufficient vegetation recovery within the linear clearings, where a lack of ground cover may increase predation risk, even for this disturbance-tolerant species (Darling et al. 2019; Pattison et al. 2020). This mechanism may be particularly relevant for the power lines sampled in our study, where vegetation remains in an early-successional state and is also subject to higher levels of human disturbance, including recreational use. These combined factors might explain the unexpected pattern of avoidance we observed, highlighting that while deer mice can tolerate disturbance, the specific conditions within linear features, particularly limited vegetation cover and heightened exposure, may offset the benefits typically associated with early-successional habitats.

For short-tailed weasels, we observed a negative response to linear feature width, contrary to our expectation that as a generalist species, they would show little sensitivity to width. This response may be linked to trophic dependencies. Short-tailed weasels primarily prey on small mammals, particularly voles and shrews, which were themselves negatively associated with wider linear features. As such, the avoidance of wider features could be partly driven by prey availability, with forest interiors and narrower seismic lines supporting higher densities of preferred prey species (Moses and Boutin 2001). The preference for narrower features and intact forest sites may also reflect the greater structural cover provided in these environments. Weasels are vulnerable to predation from raptors and larger carnivores, particularly in open environments where overhead cover is limited (Powell 1973; Korpimäki et al. 1989). Wider seismic lines and linear features often maintain low vegetation height due to ongoing vegetation management, leaving weasels exposed to aerial predators and reducing the number of suitable denning and escape sites (Linnell et al. 2017).

Overall, the expected pattern of turnover from forest-affiliated to early-successional species along wider features was not as clear as predicted. Instead of a turnover, we observed a general decline in species presence with increasing width, suggesting that many wider features may act more as low-quality habitats than actively preferred open habitats. This contrasts with previous research (Johnson et al. 1979; Shishikin et al. 2016; Darling et al. 2019), which found that while forest-dwelling species decline in wider linear features, early-successional ones increase. Our findings suggest that wider features such as pipelines and powerlines right-of-way seem to reduce local species diversity and habitat use, potentially affecting other levels of the food chain. As we used log-width as a response variable in our analysis, this suggests an exponential decline in species use as linear features get wider. In contrast, narrow (low-impact) seismic lines (3–5 m wide) seem to have higher use compared to wider seismic lines, though still lower than the forest interior.

Although our analyses detected statistically significant effects of linear feature width on small mammal use, the variance explained, particularly at the community level, was relatively low. This is not uncommon in ecological field studies, especially those involving community data and highlights the presence of unmodelled factors that could potentially affect small mammal habitat use. Additionally, we stress that our sampling design represents only a brief tem-

poral snapshot. This limited our ability to detect site-specific seasonal patterns or interannual population fluctuations, which are particularly relevant for species like meadow voles and red-backed voles, known for multi-year population cycles.

We also acknowledge that by not accounting for factors such as the presence of above-ground structures we may have oversimplified the difference between the linear features analyzed. Powerlines for example have visible metallic structures that emit electromagnetic fields, which have shown to affect the behavior of mammals (Bartzke et al. 2014) and birds (Fernie and Reynolds 2005). However, no scientific evidence supports a similar effect on small mammals, whereas many studies have shown increased small mammal abundance and diversity on powerlines (Johnson et al. 1979; Shishikin et al. 2016; Darling et al. 2019). Similarly, we excluded roads from the sampled linear features, due to their distinct structure, active vehicle use, and lack of vegetation, which would have introduced excessive variability and required different sampling methods. While this decision allowed for a focused comparison among seismic lines, pipelines, and powerlines, it limits the broader applicability of our findings. Therefore, our results should not be generalized to roads. Future research should consider the combined or comparative effects of roads and other linear disturbances on wildlife communities.

Our findings have important implications for land management. The consistent negative responses of forest-associated and generalist species to wider features suggest that approaches aimed at either restoring the linear disturbances (through replanting) or reducing their width (through policies) would likely benefit many small mammals. Moreover, our findings indicate that prioritizing pipelines and powerlines right-of-way over seismic lines may have the greatest impact, even if the lines are not fully restored across their entire width. This is because our results suggest that even if a narrow linear opening is allowed to persist on a power line right-of-way or pipeline, it wouldn't cause as strong an avoidance pattern. These conclusions align with findings from similar studies suggesting that narrow seismic lines are less likely to be perceived as habitat edges or lost habitat (e.g., Mahon et al. 2019; Tigner et al. 2015). However, we stress that focusing our efforts solely on restoring seismic lines and ignoring other disturbances like wider linear features and well sites may result in unexpected patterns of recovery by wildlife at a landscape or regional extent.

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Data availability
Raw camera trap data and location are uploaded on the WildTrax website (https://www.wildtrax.ca/home/) and may be made available upon request to the lead author at afrances@ualberta.ca.

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Supplementary material

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References

Allred, B.W., Smith, W.K., Twidwell, D., Haggerty, J.H., Running, S.W., Naugle, D.E., and Fuhlendorf, S.D. 2015. Ecosystem services lost to oil and gas in North America. Science, **348**(6233): 401–402. doi:10.1126/science.aaa4785. PMID: 25908812.

Anderson, M.J., Crist, T.O., Chase, J.M., Vellend, M., Inouye, B.D., Freestone, A.L., et al. 2011. Navigating the multiple meanings of β diversity: a roadmap for the practicing ecologist: Roadmap for beta diversity. Ecol. Lett. **14**(1): 19–28. doi:10.1111/j.1461-0248.2010.01552.x. PMID: 21070562.

Bartzke, S.G., Roel, M., Kjetil, B., Sigbjorn, S., and Eivin, R. 2014. The effects of power lines on ungulates and implications for power line routing and rights-of-way management. Int. J. Biodivers. Conserv. **6**(9): 647–662. doi:10.5897/IJBC2014.0716.

Bayne, E.M., and Dale, B.C. 2011. Effects of Energy Development on Songbirds. *In* Energy Development and Wildlife Conservation in Western North America. *Edited by* D.E. Naugle. Island Press/Center for Resource Economics, Washington, DC. pp. 95–114. doi:10.5822/978-1-61091-022-4_6.

Bennett, V.J. 2017. Effects of Road Density and Pattern on the Conservation of Species and Biodiversity. Curr. Landscape Ecol. Rep. **2**(1): 1–11. doi:10.1007/s40823-017-0020-6.

Blair, W.F. 1942. Size of Home Range and Notes on the Life History of the Woodland Deer-Mouse and Eastern Chipmunk in Northern Michigan. J. Mammal. **23**(1): 27. doi:10.2307/1374852.

Blake Hart, E., Belk, M.C., Jordan, E., and Gonzalez, M.W. 2004. Zapus princeps. Mamm. Species, **749**: 1–7. doi:10.1644/749.

Bowers, M.A., Gregario, K., Brame, C.J., Matter, S.F., and Dooley, J.L. 1996. Use of space and habitats by meadow voles at the home range, patch and landscape scales. Oecologia, **105**(1): 107–115. doi:10.1007/BF00328798. PMID: 28307129.

Brooks, M.E., Kristensen, K., Van Benthem, K.J., Magnusson, A., Berg, C.W., Nielsen, A., et al. 2017. Modeling zero-inflated count data with glmmTMB. Ecology, doi:10.1101/132753.

Canada Energy Regulator (CER). 2025. Operations and maintenance activities on pipelines regulated under the CER Act: requirements and guidance notes. Available from https://www.cer-rec.gc.ca/en/about/how-we-regulate/guidance/cera/onshore-pipelines/operations-maintenance-activities/operations-maintenance-activities-pipelines-regulated-under-cer-act-requirements-guidance-notes/operations-maintenance-activities-pipelines-guidance-notes.pdf [accessed 24 March 2025].

Carthew, S.M., Jones, K.M.W., and Lawes, M. 2013. Responses of small vertebrates to linear clearings in a South Australian woodland. Ecol. Res. **28**(6): 1003–1010. doi:10.1007/s11284-013-1082-5.

Čepelka, L., Suchomel, J., Purchart, L., and Heroldová, M. 2011. Small mammal diversity in the Beskydy Mts. forest ecosystems subject to different forms of management. Beskydy, **4**(2): 101–107. Faculty of Forestry and Wood Technology, Mendel University of Agriculture and Forestry, Division of Scientific Information (Vysoka Skola Zemedelska v Brne, Lesnicka Fakulta).

COSEWIC. 2014. COSEWIC assessment and status report on the Caribou Rangifer tarandus, Newfoundland population, Atlantic-Gaspésie population and Boreal population, in Canada. Committee on the Status of Endangered Wildlife in Canada, Ottawa. Available from https://species-registry.az.ec.gc.ca/index-en.html#/documents/7o64pGrneL9QtRiqfrkP0 [accessed 20 March 2025].

Coffin, A.W. 2007. From roadkill to road ecology: a review of the ecological effects of roads. J. Transp. Geogr. **15**(5): 396–406. doi:10.1016/j.jtrangeo.2006.11.006.

Crase, B., Liedloff, A.C., and Wintle, B.A. 2012. A new method for dealing with residual spatial autocorrelation in species distribution models. Ecography, **35**(10): 879–888. doi:10.1111/j.1600-0587.2011.07138.x.

Dabros, A., and Higgins, K.L. 2023. Vegetation recovery and edge effects of low impact seismic lines over eight-year period in boreal uplands of northern Alberta. Forest Ecol. Manag. **532**: 120850. doi:10.1016/j.foreco.2023.120850.

Dabros, A., Higgins, K.L., and Pinzon, J. 2022. Seismic line edge effects on plants, lichens and their environmental conditions in boreal peatlands of Northwest Alberta (Canada). Restor Ecol. **30**(4): e13468. doi:10.1111/rec.13468.

Dabros, A., Pyper, M., and Castilla, G. 2018. Seismic lines in the boreal and arctic ecosystems of North America: environmental impacts, challenges, and opportunities. Environ. Rev. **26**(2): 214–229. doi:10.1139/er-2017-0080.

Darling, A.F., Leston, L., and Bayne, E.M. 2019. Small-mammal abundance differs between pipelines, edges, and interior boreal forest habitat. Can. J. Zool. **97**(10): 880–894. doi:10.1139/cjz-2018-0314.

Dickie, M., Serrouya, R., DeMars, C., Cranston, J., and Boutin, S. 2017. Evaluating functional recovery of habitat for threatened woodland caribou. Ecosphere, **8**(9): e01936. doi:10.1002/ecs2.1936.

Dray, S., and Dufour, A.-B. 2007. The **ade4** Package: Implementing the Duality Diagram for Ecologists. J. Stat. Softw. **22**(4):. doi:10.18637/jss.v022.i04.

Dray, S., and Legendre, P. 2008. Testing the Species Traits–Environment Relationships: the Fourth-Corner Problem Revisited. Ecology, **89**(12): 3400–3412. doi:10.1890/08-0349.1. PMID: 19137946.

Fahrig, L. 2003. Effects of Habitat Fragmentation on Biodiversity. Annu. Rev. Ecol. Evol. Syst. **34**(1): 487–515. doi:10.1146/annurev.ecolsys.34.011802.132419.

Fauteux, D., Imbeau, L., Drapeau, P., and Mazerolle, M.J. 2012. Small mammal responses to coarse woody debris distribution at different spatial scales in managed and unmanaged boreal forests. Forest Ecol. Manag. **266**: 194–205. doi:10.1016/j.foreco.2011.11.020.

Fauteux, D., Mazerolle, M.J., Imbeau, L., and Drapeau, P. 2013. Site occupancy and spatial co-occurrence of boreal small mammals are favoured by late-decay woody debris. Can. J. For. Res. **43**(5): 419–427. doi:10.1139/cjfr-2012-0397.

Fernie, K.J., and Reynolds, S.J. 2005. The effects of electromagnetic fields from power lines on avian reproductive biology and physiology: a review. J. Toxicol. Environ. Health, Part B, **8**(2): 127–140. doi:10.1080/10937400590909022.

Fisher, J.T., and Wilkinson, L. 2005. The response of mammals to forest fire and timber harvest in the North American boreal forest. Mamm. Rev. **35**(1): 51–81. doi:10.1111/j.1365-2907.2005.00053.x.

Fortin, C. 2008. Micromammifères et emprises de lignes de transport d'énergie électrique situées en milieu forestier. Le Nat. Can. **132**(1): 32 [Verlag nicht ermittelbar].

Franklin, C.M.A., Filicetti, A.T., and Nielsen, S.E. 2021. Seismic line width and orientation influence microclimatic forest edge gradients and tree regeneration. Forest Ecol. Manag. **492**: 119216. doi:10.1016/j.foreco.2021.119216.

Fuller, A.K., Harrison, D.J., and Lachowski, H.J. 2004. Stand scale effects of partial harvesting and clearcutting on small mammals and forest structure. Forest Ecol. Manag. **191**(1–3): 373–386. doi:10.1016/j.foreco.2004.01.014.

Galindo, C., and Krebs, C.J. 1985. Habitat use and abundance of deer mice: interactions with meadow voles and red-backed voles. Can. J. Zool. **63**(8): 1870–1879. doi:10.1139/z85-278.

Getz, L.L. 1961. Home Ranges, Territoriality, and Movement of the Meadow Vole. J. Mammal. **42**(1): 24. doi:10.2307/1377237.

Glen, A.S., Cockburn, S., Nichols, M., Ekanayake, J., and Warburton, B. 2013. Optimising Camera Traps for Monitoring Small Mammals. PLoS One, **8**(6): e67940. doi:10.1371/journal.pone.0067940. PMID: 23840790.

Goheen, J.R., and Swihart, R.K. 2005. Resource selection and predation of north american red squirrels in deciduous forest fragments. J. Mammal. **86**(1): 22–28. doi:10.1644/1545-1542(2005)086%3C0022:RSAPON%3E2.0.CO;2.

Government of Alberta. 2017. 2017 Alberta's leased oil sands area files. Available from https://open.alberta.ca/publications/alberta-s-leased-oil-sands-area-map [accessed: 29 Sept 2025].

Government of Alberta. 2018a. Reclamation practices and criteria for powerlines. Edmonton: Alberta Environment and Parks. Available from https://open.alberta.ca/publications/reclamation-practices-and-criteria-for-powerlines [accessed 24 March 2025].

Government of Alberta. 2018b. Pipelines in Alberta: what landowners need to know. Agdex, 878–874. Farmers' Advocate Office, Edmonton.

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Available from https://www1.agric.gov.ab.ca/$department/deptdocs.nsf/all/agdex1125/$file/878-4.pdf?OpenElement [accessed 24 March 2025].

Government of Alberta. 2023. Pipeline Rules, Alberta Regulation 125/2023. Edmonton: King's Printer. Available from https://open.alberta.ca/publications/2023_125#summary [accessed 24 March 2025].

Government of Canada. 2019. Canadian Energy Regulator Act S.C. 2019, c. 28, s. 10. Available from https://laws-lois.justice.gc.ca/eng/acts/C-15.1/FullText.html [accessed 24 March 2025].

Gracanin, A., Gracanin, V., and Mikac, K.M. 2019. The selfie trap: a novel camera trap design for accurate small mammal identification. Ecol. Manage. Restor. **20**(2): 156–158. doi:10.1111/emr.12345.

Gregoire, J.M., Hedley, R.W., and Bayne, E.M. 2022. Canada Warbler response to vegetation structure on regenerating seismic lines. Avian Conserv. Ecol. **17**(2): art26. doi:10.5751/ACE-02262-170226.

Johnson, W.C., Schreiber, R.K., and Burgess, R.L. 1979. Diversity of Small Mammals in a Powerline Right-of-way and Adjacent Forest in East Tennessee. Am. Midl. Nat. **101**(1): 231. doi:10.2307/2424918.

Kalukapuge, T., Leston, L., Martínez-Lanfranco, J.A., and Bayne, E. 2024. Response of boreal songbird communities to the width of linear features created by the energy sector in Alberta, Canada. Avian Conserv. Ecol. **19**(2): art14. doi:10.5751/ACE-02730-190214.

Kelly, M.J., and Holub, E.L. 2008. Camera Trapping of Carnivores: Trap Success Among Camera Types and Across Species, and Habitat Selection by Species, on Salt Pond Mountain, Giles County, Virginia. Northeast. Nat. **15**(2): 249–262. doi:10.1656/1092-6194(2008)15%5B249:CTOCTS%5D2.0.CO;2.

Kirkland, G.L. 1990. Patterns of Initial Small Mammal Community Change after Clearcutting of Temperate North American Forests. Oikos, **59**(3): 313. doi:10.2307/3545141.

Korpimäki, E., Norrdahl, K., and Korpimaki, E. 1989. Avian Predation on Mustelids in Europe 1: Occurrence and Effects on Body Size Variation and Life Traits. Oikos, **55**(2): 205. doi:10.2307/3565424.

Latham, A.D.M., Latham, M.C., Boyce, M.S., and Boutin, S. 2011. Movement responses by wolves to industrial linear features and their effect on woodland caribou in northeastern Alberta. Ecol. Appl. **21**(8): 2854–2865. doi:10.1890/11-0666.1.

Laurance, W.F., Goosem, M., and Laurance, S.G.W. 2009. Impacts of roads and linear clearings on tropical forests. Trends Ecol. Evol. **24**(12): 659–669. doi:10.1016/j.tree.2009.06.009. PMID: 19748151.

Lee, P., and Boutin, S. 2006. Persistence and developmental transition of wide seismic lines in the western Boreal Plains of Canada. J. Environ. Manage. **78**(3): 240–250. doi:10.1016/j.jenvman.2005.03.016. PMID: 16112795.

Legendre, P., and Gallagher, E.D. 2001. Ecologically meaningful transformations for ordination of species data. Oecologia, **129**(2): 271–280. doi:10.1007/s004420100716. PMID: 28547606.

Legendre, P., and Legendre, L. 2012. Numerical EcologyVol. 24, 3rd ed. Elsevier, Oxford, UK.

Lemaitre, J., Fortin, D., Morris, D. W., and Darveau, M. 2010. Deer mice mediate red-backed vole behaviour and abundance along a gradient of habitat alteration. Evol. Ecol. Res. **12**(2): 203–216.

Lin, Y.-T.K., and Batzli, G.O. 2001. The Influence Of Habitat Quality On Dispersal, Demography, And Population Dynamics Of Voles. Ecol. Monogr. **71**(2): 245–275. doi:10.1890/0012-9615(2001)071%5B0245:TIOHQO%5D2.0.CO;2.

Linnell, M.A., Epps, C.W., Forsman, E.D., and Zielinski, W.J. 2017. Survival and Predation of Weasels (*Mustela erminea, Mustela frenata*) in North America. Northwest Sci. **91**(1): 15–26. doi:10.3955/046.091.0104.

Machtans, C.S. 2006. Songbird response to seismic lines in the western boreal forest: a manipulative experiment. Can. J. Zool. **84**(10): 1421–1430. doi:10.1139/z06-134.

Mahon, C.L., Holloway, G.L., Bayne, E.M., and Toms, J.D. 2019. Additive and interactive cumulative effects on boreal landbirds: winners and losers in a multi-stressor landscape. Ecol. Appl. **29**(5): e01895. doi:10.1002/eap.1895. PMID: 31121076.

Martell, A.M. 1983. Demography of southern red-backed voles (*Clethrionomys gapperi*) and deer mice (*Peromyscus maniculatus*) after logging in north-central Ontario. Can. J. Zool. **61**(5): 958–969. doi:10.1139/z83-129.

Michał, B., and Rafał, Z. 2014. Responses of small mammals to clear-cutting in temperate and boreal forests of Europe: a meta-analysis and review. Eur. J. Forest Res. **133**(1): 1–11. doi:10.1007/s10342-013-0726-x.

Morris, R.F. 1955. Population Studies on Some Small Forest Mammals in Eastern Canada. J. Mammal. **36**(1): 21. doi:10.2307/1375718.

Moses, R.A., and Boutin, S. 2001. The influence of clear-cut logging and residual leave material on small mammal populations in aspen-dominated boreal mixedwoods. Can. J. For. Res. **31**(3): 483–495. doi:10.1139/x00-186.

Oatway, M.L., and Morris, D.W. 2007. Do animals select habitat at small or large scales? An experiment with meadow voles (Microtus pennsylvanicus). Can. J. Zool. **85**(4): 479–487. doi:10.1139/Z07-023.

Oksanen, J. 2009. Ordination and analysis of dissimilarities: tutorial with R and Vegan. University Tennessee: Knoxville, TN, USA.

Oksanen, J., Blanchet, F.G., Kindt, R., Legendre, P., Minchin, P.R., O'hara, R.B., et al. 2013. vegan: Community Ecology Package. doi:10.32614/CRAN.package.vegan.

Panzacchi, M., Linnell, J.D.C., Melis, C., Odden, M., Odden, J., Gorini, L., and Andersen, R. 2010. Effect of land-use on small mammal abundance and diversity in a forest–farmland mosaic landscape in southeastern Norway. Forest Ecol. Manag. **259**(8): 1536–1545. doi:10.1016/j.foreco.2010.01.030.

Pattison, C.A., Castley, J.G., and Catterall, C.P. 2020. Seismic linear clearings alter mammal abundance and community composition in boreal forest landscapes. Forest Ecol. Manag. **462**: 117936. doi:10.1016/j.foreco.2020.117936.

Pattison, C.A., Quinn, M.S., Dale, P., and Catterall, C.P. 2016. The Landscape Impact of Linear Seismic Clearings for Oil and Gas Development in Boreal Forest. Northwest Sci. **90**(3): 340. doi:10.3955/046.090.0312.

Powell, R.A. 1973. A Model for Raptor Predation on Weasels. J. Mammal. **54**(1): 259–263. doi:10.2307/1378888.

Ramirez, P., and Hornocker, M. 1981. Small Mammal Populations in Different-Aged Clearcuts in Northwestern Montana. J. Mammal. **62**(2): 400–403. doi:10.2307/1380724.

Riege, D.A. 1991. Habitat Specialization and Social Factors in Distribution of Red and Gray Squirrels. J. Mammal. **72**(1): 152–162. doi:10.2307/1381990.

Ries, L., Fletcher, R.J., Battin, J., and Sisk, T.D. 2004. Ecological Responses to Habitat Edges: Mechanisms, Models, and Variability Explained. Annu. Rev. Ecol. Evol. Syst. **35**(1): 491–522. doi:10.1146/annurev.ecolsys.35.112202.130148.

Riva, F., Acorn, J.H., and Nielsen, S.E. 2018. Localized disturbances from oil sands developments increase butterfly diversity and abundance in Alberta's boreal forests. Biol. Conserv. **217**: 173–180. doi:10.1016/j.biocon.2017.10.022.

Shishikin, A., Timoshkin, V., Gurov, A., Ekimov, E., Egunova, M., Loshchev, S., and Astapenko, S. 2016. Animal populations on the glades of electric power line-500 kw in fir and pine formations of the southern taiga. Sibirskij Lesnoj ZurnalSiberian Journal of Forest Science(No.2): 59–70. Russian Academy of Sciences, Siberian Branch Publishing House.

Sikes, R.S., and Gannon, W.L. 2011. Guidelines of the American Society of Mammalogists for the use of wild mammals in research. J. Mammal. **92**(1): 235–253. doi:10.1644/10-MAMM-F-355.1.

Soria, C.D., Pacifici, M., Di Marco, M., Stephen, S.M., and Rondinini, C. 2021. COMBINE: a coalesced mammal database of intrinsic and extrinsic traits. Ecology, **102**(6): e03344. doi:10.1002/ecy.3344.

Statistics Canada. 2023. 2021 Census Boundary files. Available from https://www12.statcan.gc.ca/census-recensement/2021/geo/sip-pis/boundary-limites/index2021-eng.cfm?Year=21 [accessed 29 September 2025].

Stern, E.R., Riva, F., and Nielsen, S.E. 2018. Effects of Narrow Linear Disturbances on Light and Wind Patterns in Fragmented Boreal Forests in Northeastern Alberta. Forests, **9**(8): 486. doi:10.3390/f9080486.

Storm, J.J., and Choate, J.R. 2012. Structure and Movements of a Community of Small Mammals Along a Powerline Right-Of-Way in Subalpine Coniferous Forest. Southwest. Nat. **57**(4): 385–392. doi:10.1894/0038-4909-57.4.385.

Sullivan, T.P., Lautenschlager, R.A., and Wagner, R.G. 1999. Clearcutting and burning of northern spruce-fir forests: implications for small

mammal communities. J. Appl. Ecol. **36**(3): 327–344. doi:10.1046/j.1365-2664.1999.00408.x.

Suzuki, N., and Hayes, J.P. 2003. Effects of Thinning on Small Mammals in Oregon Coastal Forests. J. Wildli. Manage. **67**(2): 352. doi:10.2307/3802777.

Sullivan, T.P., Sullivan, D.S., and Sullivan, J.H., 2017. Mammalian responses to windrows of woody debris on clearcuts: abundance and diversity of forest-floor small mammals and presence of small mustelids. Forest Ecology and Management **399**:143–154. doi:10.1016/j.foreco.2017.05.028.

Ter Braak, C.J.F., Cormont, A., and Dray, S. 2012. Improved testing of species traits–environment relationships in the fourth-corner problem. Ecology, **93**(7): 1525–1526. doi:10.1890/12-0126.1.

Tigner, J., Bayne, E.M., and Boutin, S. 2015. American Marten Respond to Seismic Lines in Northern Canada at Two Spatial Scales. PLoS One, **10**(3): e0118720. doi:10.1371/journal.pone.0118720.

Tisell, H.B. 2018. Home range and microhabitat associations of the southern red-backed vole (*Myodes gapperi*) in New Hampshire forests. M.Sc. thesis, Department of Biological Sciences, University of New Hampshire, Durham, NH.

Tisell, H.B., Degrassi, A.L., Stephens, R.B., and Rowe, R.J. 2023. Southern Red-backed Vole (*Myodes gapperi*) habitat associations in northern New England forests. J. Mammal. **104**(6): 1400–1407. doi:10.1093/jmammal/gyad068.

Van Rensen, C.K., Nielsen, S.E., White, B., Vinge, T., and Lieffers, V.J. 2015. Natural regeneration of forest vegetation on legacy seismic lines in boreal habitats in Alberta's oil sands region. Biol. Conserv. **184**: 127–135. doi:10.1016/j.biocon.2015.01.020.

Vanderwel, M.C., Malcolm, J.R., Caspersen, J.P., and Newman, M.A. 2010. Fine-scale habitat associations of red-backed voles in boreal mixedwood stands. J. Wildl. Manag. **74**(7): 1492–1501. doi:10.1111/j.1937-2817.2010.tb01276.x.

Venier, L.A., Thompson, I.D., Fleming, R., Malcolm, J., Aubin, I., Trofymow, J.A., et al. 2014. Effects of natural resource development on the terrestrial biodiversity of Canadian boreal forests. Environ. Rev. **22**(4): 457–490. doi:10.1139/er-2013-0075.

Wood, S. 2017. Mixed GAM computation vehicle with GCV/AIC/REML smoothness estimation. R package version. Vol. 1. 8. Available from https://cir.nii.ac.jp/crid/1370285710777568774.

Wrigley, R.E., Dubois, J.E., and Copland, H.W.R. 1979. Habitat, Abundance, and Distribution of Six Species of Shrews in Manitoba. J. Mammal. **60**(3): 505–520. doi:10.2307/1380091.

Zwolak, R. 2009. A meta-analysis of the effects of wildfire, clearcutting, and partial harvest on the abundance of North American small mammals. Forest Ecol. Manag. **258**(5): 539–545. doi:10.1016/j.foreco.2009.05.033.

Zwolak, R., and Foresman, K.R. 2007. Effects of a stand-replacing fire on small-mammal communities in montane forest. Can. J. Zool. **85**(7): 815–822. doi:10.1139/Z07-065.

Can. J. Zool. Downloaded from cdnsciencepub.com by 162.106.4.157 on 10/23/26